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# Unveiling the Missing Innovation - An Actor-Network Theory Analysis of Regulatory Efforts Towards Universal Design

## Abstract

Purpose

The purpose of this paper is to study and analyse challenges in a regulatory innovation process the field of accessibility and Universal Design (UD) practice. The study takes it starting point in a Danish regulatory initiative concerning innovation in the building regulations. The aim of the initiative was to facilitate a new approach, which acknowledges human diversity to a greater degree than is the case with the traditional accessibility approach that relies on prescriptive requirements.

Design/methodology/approach

The Actor-Network Theory perspective of translation is employed to analyse the process and the complexity of the network of elements involved in the efforts to develop a new approach. The analysis is grounded in a qualitative survey, including semi-structured interviews, with respondents from architectural firms and respondents from local building authorities (municipalities) of their experiences with the building regulations and their way of working with accessibility or UD. The analysis concerned the following questions: 1) Which actors were intended to be enrolled in the network of an initiative about UD innovation? and 2) What factors contributed to the failure of the efforts of the building sector to develop innovative UD solutions?

Findings

The findings show that none of the actors involved succeeded with an innovation solution. Lack of knowledge about UD, accessibility, and the low status of the subject together with tight economy and the anticipation of expensive solutions, prevented the architectural firms and building surveyors from embracing innovative solutions. Based on the findings, the discussions reflect on the implications for future adoption of UD in future policy making.

Originality

Looking into the complexity of an innovative initiative, this paper contributes to the field of regulation concerning UD in architecture.

Keywords: Accessibility, Actor-Network Theory, Architecture, Architectural practice, Barriers, Innovation, Legislation, Policy, Socio-technical systems, Translations, Universal design

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## 1. Introduction

Universal Design (UD) represents an innovative way of thinking about the built environment; it is, so to speak, a process of innovation (Steinfeld, 2013) while accessibility is, and has been, about regulation and compliance without acknowledging human diversity (Ruptash, 2013; Steinfeld, 2013; Salmen, 2011). As early as 2010, Steinfeld (2010) points at a barrier towards accelerating the pace of the adoption of UD practices. This involves hastening the shift from mere accessibility to a fully comprehensive UD approach. UD contributes with spatial facilitators, as opposed to accessibility, which focuses on removing barriers (Steinfeld, 2010; Steinfeld and Tauke, 2002). Nevertheless, there are different understandings of UD at play in the building sector; some see it in the same way as accessibility (Salmen, 2011, Steinfeld and Maisel, 2012), others argue that it only addresses people with disabilities (Yosof and Jones, 2014), and other again look at it as a concept about social sustainability and architectural quality (Grangaard and Lygum, 2022). While some associate UD with innovation, they find it hard and too abstract to use (Erdtman *et al*, 2021).

Despite a growing interest in UD, the adoption of UD in the building sector has been slow. The literature has pointed to numerous reasons. Some revolve around prejudices on the economy (Zallio and Clarkson, 2021) and the lack of business cases and economic incentives (Heylighen *et al*, 2017). Other reasons include a lack of knowledge and competences (Grangaard, 2018a; Heylighen *et al*, 2017; Grangaard, 2016), insufficient integration in the curriculum of architecture schools (Heylighen *et al*, 2017), a peripheral position on the design agenda and its absence in design awards (ibid.), and that is not seen as a policy tool in architectural policies (Grangaard, 2018b). The missing priority and lack of focus from clients and architectural firms are other reasons (Grangaard, 2016; Grangaard, 2018a; Ielegems *et al*, 2019). Traditionally, clients and architectural firms have held narrow perceptions of the inclusive dimension of architecture, often linking it solely with sectors such as healthcare, exemplified by hospitals and assisted living facilities (Van der Linden *et al*, 2016) or primarily public buildings (Müller *et al*, 2022). A recurring theme in the literature is that regulations are perceived as conflicting requirements that restrict creativity (Heylighen *et al*, 2017; Gray *et al*, 2003). Moreover, practitioners often interpret regulatory minimum-level requirements for accessibility as maximum-level requirements and lack the knowledge to rise above this level in their solutions (Steinfeld, 2023; Steinfeld, 2010). Extant research has found that laws and regulations are not sufficient to foster a

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transformation to UD because such a change requires an acknowledgement of the combination of regulation and knowledge (Steinfeld, 2010) as well as advocacy (Ruptash, 2013). Nevertheless, a change in the approach to regulation is necessary (Steinfeld, 2010). Accordingly, Lid (2013) argues that UD should be understood and practiced simultaneously on several levels to succeed. Values and laws operate at the macro level, whereas technical standards serve as instruments at the meso level, resulting in people´s experience of UD through an accessible and usable environment at the micro level.

While the literature acknowledges the complexity of regulation (Imrie and Street, 2010), also in relation to the adoption of UD (Steinfeld and Maisel, 2012), it has, with a few exceptions (Imrie, 2012), tended to focus on the potential if regulations are developed to fit architectural practice (Van der Linden *et al*, 2016). Architects request less prescriptive requirements (Wauters et al, 2014), just as there is an interest among architectural firms for greater room for manoeuvre (Gramkow *et al*, 2022; Grangaard, 2016) and for performance-based requirements (Grangaard, 2016; Kirkeby *et al*, 2014). An introduction of performance-based requirements will fundamentally transform the existing regulatory system that has black boxed accessibility, being opaque to outsiders and being taken for granted (Grangaard & Gottlieb, 2019). In a situation of change, Meacham *et al* (2017) argue that studying and understanding a regulatory system requires a socio-technical system framework to identify interactions between the different elements and actors within the system. The Actor-Network Theory (ANT) perspective of translation (Callon, 1986) helps us to go beyond what is taken-for-granted in policymaking and regulation by studying the complexity of the network of elements and actors that must cooperate to succeed.

On this basis, the focus of this paper is to study the processes and problems that arise when regulating by asking the research question: 1) Which actors were intended to be enrolled in the network of an initiative about UD innovation? And 2) What factors contributed to the failure of the efforts of getting the building sector to develop innovative UD solutions? Drawing on a Danish case of an initiative about UD innovation in a new version of the buildings regulations from 2018 (BR18), this paper presents the translation process of this initiative and its implications. In the light of future policy making, we will discuss how to support innovation in a regulatory system in the field of UD. We have structured the remaining paper into five parts. The theoretical framework of translation is presented in section 2. Section 3 introduces the context and the

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method. In section 4, we present the findings. Then follows the discussion in section 5 and the conclusion in section 6.

## 2. The theoretical framework

In this paper, we are not interested in the designed solutions but the network with its processes, roles, and relations that arise and change when a regulatory framework is transformed. ANT offers a perspective for examining the interactions and networks among these heterogeneous elements. It originates from Science and Technology Studies (STS) and studies of laboratories. In these studies, the construction of scientific facts was followed and mapped in their interplay with a network of measurements instruments and apparatuses, showing that scientific results are created in contingent socio-material practices. (Latour and Woolgar, 1986) Thus, the laboratory studies confronted the traditional understanding of nature and society as separate or independently existing domains, e.g., the social does not exist as a thing but between things as ‘a type of connection’ (Latour, 2005).

As Law points out, ANT encompasses a set of material-semiotic tools and analytical methods. It approaches entities in both social and natural realms as outcomes continually shaped by the intricate webs of relations in which they exist (Law, 2008). Within ANT, all entities are considered outcomes of diverse networks where individuals, technologies, and other elements are interlinked and exert influence on one another. ANT thus emphasizes the agency not only of humans but also of non-human elements in shaping relations. Non-human elements can be buildings and artefacts in space that can contribute to the regulation of human behavior, and human actions can be distributed to physical elements (Latour, 1992).

Regarding the changes made to BR18, prescriptive requirements outline precise instructions for the design, construction, and procurement of specific building components. On the other hand, performance-based requirements delineate and detail functional characteristics that can be achieved through various approaches in the design phase. Instead of viewing such requirements solely as technical documents, we perceive them as strategic assets that inscribe associations and programmes of action. They actively contribute to shaping interactions among actors, objects, instruments, and other elements within the actor-network (Akrish *et al*., 2002) that constitutes a building project.

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### *2.1 Translation*

In ANT, translation is a process that must be understood as both a translation but also as mediation, and a relation between actors in a network (Latour, 2005). It occurs by negotiation and the delimitation of different actors' identity, interests, interactions, and possibilities for maneuvering, attempting to lock them into place (Callon, 1986). Callon outlines four moments or stages of translation that provide a framework for studying power and understanding the dynamic process through which actors and elements are enrolled, aligned, and stabilised within an actor-network when a new technology is introduced. It emphasises the contingent and negotiated nature of relationships in social and technological contexts. Power is not given a priori but is a result of interactions between actors. Thus, we can use his model of transition to recognise and study the complexity of regulation.

The initial moment about *problematisation* involves framing a problem by a primary actor who wants to create a change. Actors are identified, and their interests aligned with network objectives, framing issues to engage them and make themselves indispensable. The concept of obligatory passage point (OPP) signifies commitment and alignment with network goals, crucial for stabilisation and functioning because every actor find an interest in the OPP and accept it (Callon, 1986).

*Interessement*, the second movement, involves an entity asserting and solidifying others' identities through problematisation, using various mechanisms. To prevent actors from engaging in alliances elsewhere, the problematising entity intervenes by constructing devices to connect them exclusively to its network, competing with external actors and elements. These devices will have different characters because they must compete with different external actors (Callon, 1986).

The third moment is *enrolment* where the actors become interested in the goals of the network, accept their roles in it, and are convinced to participate. This process often involves distribution of roles, negotiation e.g. of the actor's identity, persuasion, and the construction of narratives or discourses that make the proposed network or system appealing to potential participants. Alliances with other actors and elements can hinder the enrollment at same time as a network requires support from other actors to be strengthened (Callon, 1986).

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The final moment *mobilisation* happens when the actors of the network are enrolled, and external elements and actors support it. Mobilisation makes the network stabilized, and the interactions are made predictable. Nevertheless, controversy can lead to actors withdrawing their support or losing external allies – such actions are frequently viewed as acts of betrayal (Callon, 1986).

Building on this foundation, our objective is to unravel and understand why the possibility of designing innovative solutions in the field of UD has not gain a hearing in the building sector. We use the model of translations to record the actors and interactions between the elements and actors in the network around BR18.

## 3. Method

This part of the paper introduces the research context, the research design, and finally, the data analysis.

### *3.1 The Danish regulation context*

A transformation towards performance-based requirements is underway in many countries (Meacham et al, 2005). In Denmark too, such a transformation has taken place in relation to fire safety, statics, and energy consumption. In general, the progressions since the early 1990s have been shaped within a framework of neoliberal governmentality, characterised by an emphasis on deregulation and the promotion of a free market (Gottlieb and Frederiksen, 2020).

Since 1972, the Danish building regulations have had the provision for requirements ensuring that people with reduced ability to move and orient themselves can use buildings with public access, and housing for sale or hire (Byggeloven, 1972). The first prescriptive requirements were implemented in 1977 (Bygningsreglementet, 1977). In 2013, the Danish Government launched a disability policy action plan, highlighting the challenge of innovation in relation to accessibility and searching for new approaches to regulation (Regeringen, 2013). At that same time, the Danish Government aimed to strengthen the Danish building construction industry and launched a strategy with initiatives to make it easier to conduct business in the construction industry. One focus area was quality in regulation with the objective of modernising the

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regulatory system to support growth, productivity, and innovation through simplification of the regulation and effective handling of building permits (Regeringen, 2014).

In general, this strategy caused a comprehensive transformation of the regulatory system manifested in new building regulations describing new procedures and requirements (Building Regulations, 2018). Regarding accessibility, the local authority would no longer control the application for building permit. Instead, the applicants should document the projects as built in relation to the notification of completion and occupation permit. On that basis, the local authority carries out a random sample check of every tenth occupation permit to control if the documentation complies with the requirements in the building regulations. In the field of fire safety and statics, the handling of control was taken over by external certified consultants (ibid.).

### *3.2 BR18*

Simplification was the intention of BR18. However, BR18 consists of 22 thematic chapters that supersede the eight chapters of buildings regulations 2015 (BR15). Interestingly, accessibility as a word is literally erased from BR18. ‘Chapter 2 - Access’ and ‘Chapter 9 - Building layout’ have objects clauses that emphasise the users – even though BR18 is still built around prescriptive requirements and specifies that these requirements must be met.

In 2007, Denmark ratified the Convention on the Rights of Persons with Disabilities (CRPD) which, in relation to the built environment, operates with the concept of UD and designing for everybody regardless of disability. The concept of UD is not implemented in the requirements of BR18 but is exclusively introduced in a ‘guideline on the concept of users’. This guideline advises to think in terms of UD to encompass human diversity and not just mobility and visual impairments addressed by the requirements of BR18. The guideline stresses the importance of ensuring equal access and building layouts for all users and suggests considering alternative solutions to supplement established measures when necessary. If these solutions introduce new methods, e.g., for access conditions, dispensation may be required. To navigate such situations, a preliminary dialogue meeting with the municipality is recommended. This approach, facilitated by the client or architectural firm, ensures regulatory compliance while accommodating diverse user needs (Guideline BR18).

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### *3.3 Research design*

Follow the actors is the mantra in ANT (Latour, 1987) and based on semi-structured single interviews with respondents from architectural firms and respondents from local building authorities (municipalities), in this study we follow the actors. The interviews were conducted in the autumn of 2023. The architectural firms were identified based on an information-based strategy (Flyvbjerg, 2006) focusing on variation on one dimension. We, therefore, contacted two kinds of architectural firms. The first, we classified as having a special focus on accessibility and UD in the expectation that if some architectural firms work with innovative solutions, we could find them here. The second type of architectural firms consisted of firms where it was not expected that they had a special focus. However, we would not rule out the possibility that they have worked with innovative solutions. We interviewed 12 architectural firm, six with assumes focus and six without assumed focus on accessibility and UD (see Table 1). The selection of companies was based on our knowledge about the field obtained from our teaching at a professional master's programme at the university and as editor at a website about UD. The firms were distributed across a breadth of sizes.

| **Informant** | **Size of the firm** | **Type of architectural firm** | **Focus on accessibility and UD** | **Not a focus on accessibility and UD** |
|---|---|---|---|---|
| Technical designer | International; 204 employees in Denmark | Buildings Landscape & urbanism | | x |
| Architect and project director | 145 employees | Buildings Landscape & urbanism | | x |
| Architect, design manager | 63 employees | Buildings | | x |
| Architect, partner | 11 employes | Buildings | x | |

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| | | | | |
|---|---|---|---|---|
| Architect, partner | 8 employees | Buildings | x | |
| Architect, partner | 43 employees | Buildings | x | |
| Constructing Architects | International; 410 employees in Denmark | Buildings Landscape & urbanism | x | |
| Senior landscape architect | International; 302 employees in Denmark | Buildings Landscape & urbanism | | x |
| Landscape architect | 33 employees | Landscape Urbanism | x | |
| Landscape architect | 33 employees | Landscape | | x |
| Landscape architect | 58 employees | Landscape Urbanism | | x |

Table 1 – An overview of the architectural firms

# Table I – An overview of the interviewed architectural firms (Source: Table created by authors).

The municipalities were also selected based on a strategy of variation, with a focus on variation on several dimensions including geography, type (rural/urban) and size, to encompass municipalities with a wide range of applications for different kinds of buildings. In this process, we also aimed to find informants who had experience both with case handling of building permits and random sample checks according to BR18, as well as experience with earlier building regulations. To facilitate this, we sent emails to the selected municipalities well in advance, asking them to identify and choose the right informant for the task. In total, we interviewed 12 municipalities. In two of the interviews, the municipalities pointed out two employees to cover both case handing of building permits and random sample checks.

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We utilised a semi-structured interview guide for the architectural firm and for the municipalities, respectively. The interview guide for the architectural firms comprised questions about the informant's experience of BR18 and their approaches to accessibility or UD in relation to innovative solutions, dialogue meetings, and dispensation, as well as the rationale behind both traditional solutions and innovative solutions, or any barriers they encountered. Conversely, the interview guide for the municipalities focused on 1) gathering experiences from the sample checks to ascertain whether the level of accessibility in the buildings projects has been raised or not and 2) experiences from dialogue meetings with clients/architectural firms to discover the level of accessibility, and their understanding of the rationale or the barriers to innovation. The interviews lasted 30 – 60 minutes each. All quotations from the empirical material have been translated into English by the authors.

### *3.3 Data analysis*

The data analysis was conducted in two steps following an abductive approach. Notes were taken during the interviews. After each interview, the notes were categorized and recorded in two spreadsheets structured according to the two interview guides. During the first stage of analysis, we examined the impact of the BR18 in relation to dispensation, involvement in dialogue meetings, and the impression of projects and their level of accessibility or UD. We coded experienced barriers together with the informants' experience to determine the factors influencing whether the architectural firms choose to work on traditional or innovative solutions. At this step, we distinguished between the two kinds of firms and the building surveyors.

In the second stage, we coded the empirical material based on Callon´s model of translation. First, we searched the empirical material for potential entities to include in the network. Then, we examined the process of problematisation to identify goals and obstacle-problems based on the intentions of BR18 and literature about the Danish context (c.f. Grangaard, 2018a). Subsequently, we engaged in a coding process focusing on devices of interessement and competing alliances working in diagrams with headlines. Initially, we outlined all kinds of alliances for the architectural firms and the building surveyors. Then, in several iterations, we grouped them and formulated headlines that could encompass the nuances without being overly detailed. The data analysis resulted in the three figures (figure 1, 2 and 3) presented in the findings. All of these are based on the principles of Callon´s diagrams (1986).

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## 4. Findings

The findings are structured in relation to the four moments of translation included in Callon's (1986) framework. Initially, we use the theoretical framework to depict the network around the regulatory initiative and the expected expectations, roles, and actions needed to succeed – in other words, how the situation should have unfolded. Subsequently, we examine what transpired in the network, illustrating actors and elements that impeded the different entities from becoming enrolled and mobilised in the OPP due to alliances with other networks.

### *4.1 Introduction to the findings*

Nearly half of the informants from the architectural firms were aware of the existence of the guideline on the concept of users in BR18, but it did not play a significant role in their practice. One informant mentioned that the accessibility level of BR18 was already high enough. Furthermore, the guideline did not have a significant impact on the municipalities either. A building surveyor explained that she read it when it was published but have not revised it since. However, she acknowledged that the guideline made sense to create awareness of human diversity: "It is good to focus on other than wheelchair users. But we fall back and forget that it is not only is about wheelchairs." (Building surveyor). None of the architectural firms had applied for a dispensation for innovative solutions; they applied for traditional dispensation to deviate from requirements, as has always been possible. This was consistent with the fact that none of the twelve municipalities had been involved in cases with innovative solutions.

### *4.2 The network and the obligatory passage point*

Problematisation is first the moment of translation. The primary actor was the agency behind BR18, which constructed the network and framed a problem relevant to the interests and identity of entities involved as a means of aligning them and making the agency indispensable (figure 1). In addition to the architectural firms and the building surveyors, the entity of clients was identified too.

The OPP of this network is innovation in the field of accessibility and UD. Subsequently, every entity had their own interest and goal in relation to the OPP (see figure 1). The goals of the agency were innovation in relation to equality and inclusive buildings, whereas the client's goals were higher quality of the built projects. The architectural firms' goal was room for manoeuvre in relation to architectural quality. The last entity of this

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network was the building surveyors. In general, their role in relation to building permits was reduced with BR18. However, their former role as professionals and intermediaries in a powerful way when processing the applicants for building permits were re-activated in the dialogue meetings and the processing of dispensations in relation to innovative solutions. Thus, professionalism and power were the goals of the building surveyors.

Similar, an obstacle-problem for each entity existed that can block or hamper the commitment to the OPP. The obstacle-problem for the clients was the economic reality if they found innovative solutions too expensive, while the use of prescriptive requirements was an obstacle-problem threatening the architectural firms' way into the OPP. Accountability, was the obstacle-problem for the building surveyors if there was a focus on transparency in their assessment and decisions about building permits because it could result in a situation where they revert to the mindset of prescriptive requirements based on tangible measures. A relapse to the well-known prescriptive requirements would be a risk for the clients too.

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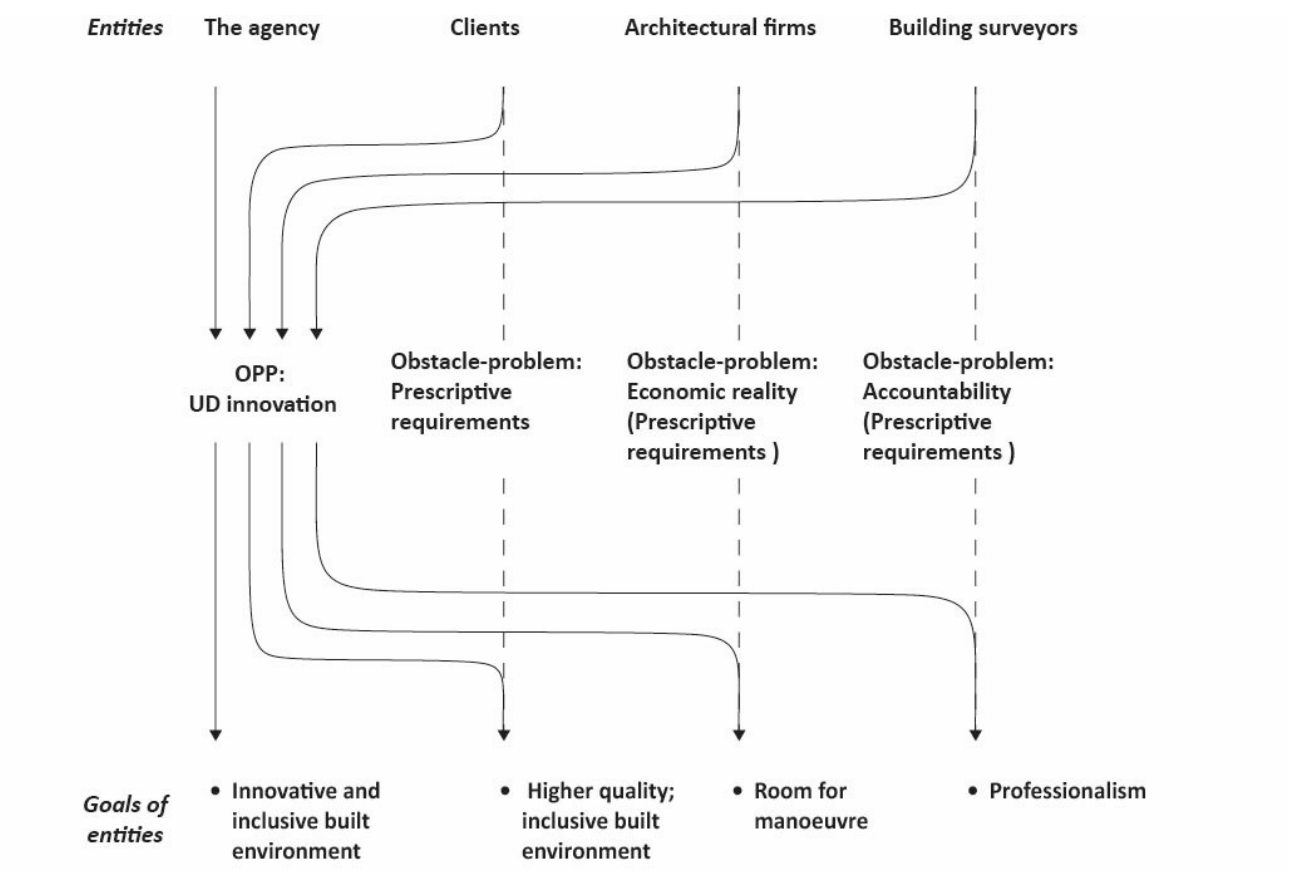



Figure 1: This figure illustrates the obligatory passage point (OPP) as being innovation and shows the various obstacles and goals of the entities expected to be enrolled in the network (Source: Figure created by authors based on Callon (1986)).

### *4.3 The devices of 'interessement'*

The second moment of translation concerns ‘interessement’. Now, we will examine the devices of 'interessement' employed by the agency to lock the actors of the network into place by supporting their practice and preventing them from allying themselves with others. At a broader level, the agency formulated objects clauses in chapter 2 and 9 in BR18, aimed at equality in access and building layout for users. Furthermore, the possibility of working in an innovative way was described in the ‘guideline on the concept of users’ in BR18. Here clients and architectural firms were encouraged to engage in dialogues meetings with the municipality, and the possibility of dispensation was mentioned. However, none of this information was presented in chapter 1, ‘Administrative provisions’.

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The agency recognised that the sector needed a knowledge boost to achieve its goals. Thus, a third initiative was the launch of a new website in 2019 designed to support a change in attitude and practice in the building sector. This website presented knowledge about UD and communicated that UD was about 1) everybody and not only people with disability, 2) architecture of high quality, and 3) value creation (Grangaard, 2021). All these devices of 'interessement' are symbolised by the grey arrow in Figure 2 and 3.

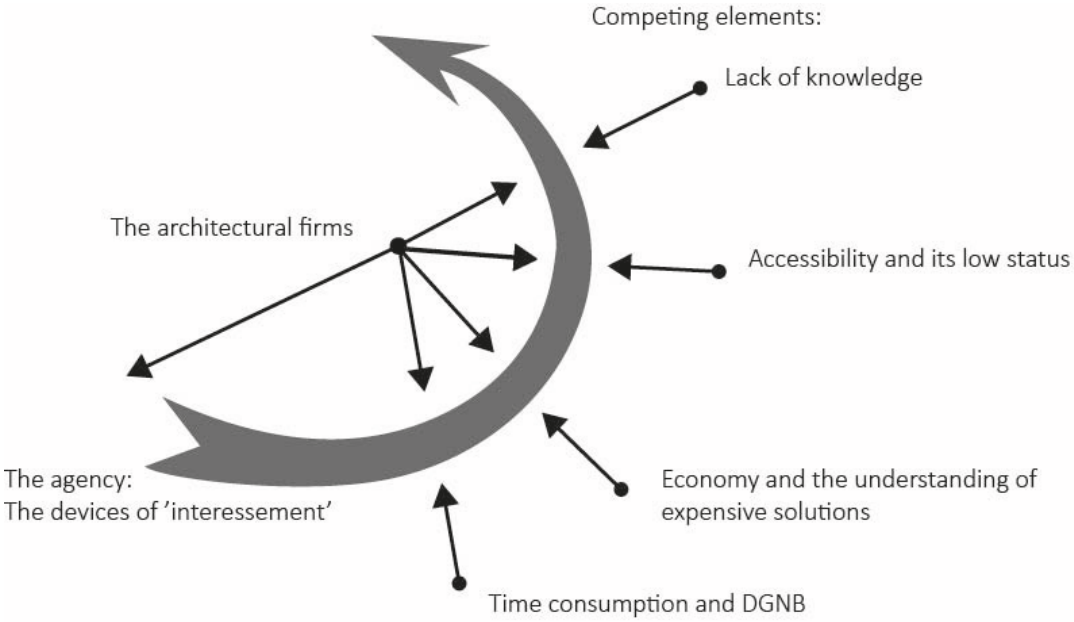



Figure 2: This figure illustrates the process of interessement for the architectural firms in relation to various competing elements. The grey arrow symbolises the devices of 'interessement' that the agency believed would prevent the firms from forming alliances with competing elements (Source: Figure created by authors based on Callon (1986)).

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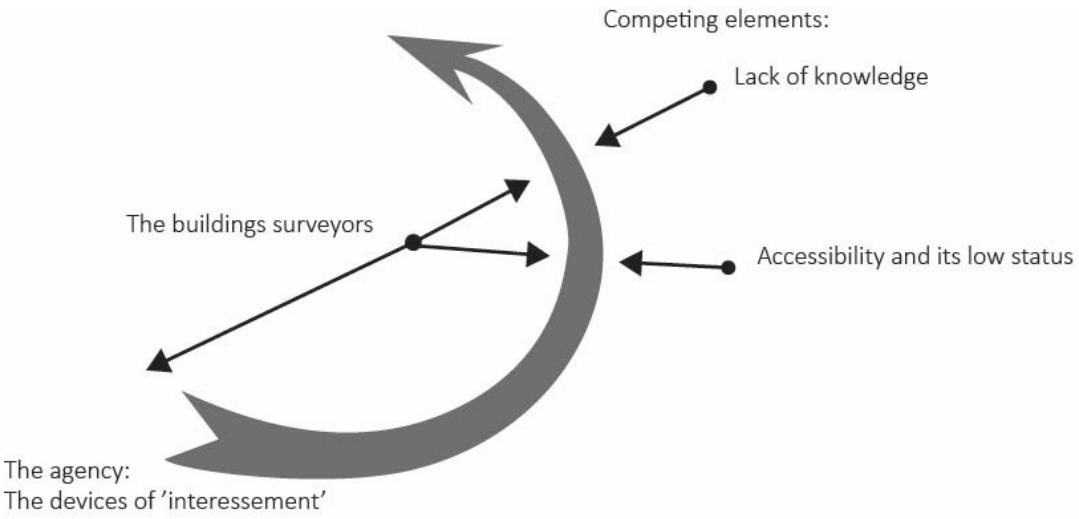



Figure 3: This figure illustrates the process of interessement for the building surveyors in relation to two competing elements. The grey arrow symbolises the devices of 'interessement' that the agency believed would prevent the surveyors from forming alliances with competing elements (Source: Figure created by authors based on Callon (1986)).

### *4.4 Enrollment to the network*

Enrollment is the third moment of translation and deals with the alignment of interests and actions of various actors towards a common goal. The interest in this network varied within every entity. Architectural firms working with innovation in areas not covered by the jurisdiction of BR18 were not interested in the network, while others lacked belief in the initiative and expected it to be too troublesome. Some architectural firms interested in UD were involved in developing innovative solutions and thus were on a trajectory towards the network. However, their involvement was short-lived because the agency failed to keep them enrolled and fully convinced that success was possible. They experienced that dialogue with the building surveyors obstructed their design proposals. Especially in a case where the client too was interested in new solutions, the mindset of the building surveyors resulted in disappointment and resignation. This forced the architectural firm to abandon their effort because the result would be that the client would only adhere to the usual prescriptive requirements in the future.

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Below, we will demonstrate how different elements or actors hindered the enrollment in the network by supporting and forming alliances to other networks. Elements related to economy, mindset, and lack of knowledge shaped these alliances towards the traditional accessibility solutions. Figure 2 and figure 3 provide an overview of these elements in relation to architectural firms and the building surveyors.

*4.4.1 An alliance with the lack of knowledge*

The architectural firms that attempted to engage in dialogue meeting found that the building surveyors only could relate to the prescribed solutions of BR18. They perceived the buildings surveyors as being hesitant to take risks due to a low competence level and fear of accusations of neglecting certain user needs. As an architect explained: "The building surveyors do not know how to deal with something other than what they have been used to for the last 15 years" (architect, architectural firm). This informant called for examples of projects with alternative solutions and examples of dialogue leading to the municipality's approval of innovative solutions.

From the building surveyor's perspective, the situation was different. None of them had been involved in dialogue or applications regarding innovative solutions. However, some building surveyors expressed willingness to consider innovative solutions but questioned the level of knowledge among architectural firms, particularly regarding user needs. One building surveyor lamented the absence of accessibility auditors involved in projects, as they improve the accessibility level without additional costs in ways that architectural firms could not achieve by themselves. One building surveyor emphasised that architectural firms are extremely familiar with the prescriptive accessibility requirements, to the extent that they would struggle if they were asked to think innovatively. Conversely, the fear of failing led to a tendency to shy away from innovation. Instead, they adhered to requirements and the instructions and check lists from the Danish Building Research Institute, opting for the safe route even when operating at a higher level of accessibility. Several building surveyors agreed that it was thought-provoking that the creativity of the applicants was primarily evident in circumventing BR18 requirements rather than proposing innovative solutions, which were viewed as attempts at cheating. At the same time, some buildings surveyors struggled to imagine any innovative solutions, as a one explained: "There are ramps, lifting platforms and elevators, what else?" (Building surveyor).

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*4.4.2 Alliances with the accessibility and its low status*

Some architectural firms found the requirements to be ambiguously formulated, leading to doubt and the need for interpretation. In some cases, it was necessary to interpret them and in other cases, it was possible to get quickly around them because accessibility was not a priority. As a result, some architectural firms had to fight for compliance when accessibility was not a priority for their clients. These firms never reached a point where considering innovative approaches was feasible. Others were not interested in developing new solutions because they believed BR18's accessibility level was sufficient or that certain requirements did not made sense.

However, many architectural firms and buildings surveyors were surprised and frustrated by the low priority placed on accessibility by clients, which often viewed it as a burden. One building surveyor explained, that accessibility as a field has a low status, like power lines and wastewater management. Moreover, accessibility was not a theme in the planning process, with district plans based on the client´s project making it more difficult to demand more accessibility than the prescriptive requirements encompassed. Furthermore, not all architectural firms and building surveyors saw and understood accessibility as an integrated part of the architecture. A building surveyor explained that they had a good dialogue also about accessibility, where BR18 did not provide discussions, but the architectural expression did, and it was believed that the client would rather spend money on the architecture. This mindset discouraged interest in developing new solutions, with few architectural firms discussing UD. Particularly, the building surveyors and the architectural firms without a focus on UD emphasised accessibility. As a result, the use of the concept accessibility functioned as an alliance that pulled the actors away from the network.

*4.4.3 Alliances with economy and the understanding of expensive solutions*

Some building surveyors pointed out that there was a notion that preparing innovative solutions was more expensive than traditional ones. However, if innovative solutions were considered from the start, they would not necessarily be more expensive. Cost was an element that affected projects, priorities, and approaches to work. It was commonly experienced that economic considerations were a determining factor in the way the architectural firms operate. This was often linked to the ambitions of most clients, who were typically satisfied with merely meeting the requirements of BR18. As one architect formulated it: "The client does not pay for more". (Architectural firm). Several architectural firms expressed surprise at the lack of ambitions from clients and their reluctance to explore more challenging options. For example, one architect was taken aback when

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a client at a special school showed little interest in a variety of toilet options that could better meet the diverse needs of users. Another architect recounted how a client only selected 70% of the wayfinding initiatives proposed in their design.

A building surveyor emphasised the client´s priority: "The client just wants to meet the requirements from the start, then they won't have trouble with it later. There are no alternative solutions. People go for what they know – then they are home safe." (Building surveyor). Another economic consideration was the lead contractor's tendency to opt for the lowest level of accessibility to save on projects costs.

#### *4.4.4 Alliances with time consumption and sustainability certification*

The aspect of time intersected economic consideration, as the saying "time is money" tells, and a new way of working required extra time. In Denmark, over the past decade, obtaining a sustainability certification for a building project has been possible. However, compliance with BR18 regulations is a prerequisite for this certification. This requirement acted as a barrier to architectural firms' engagement in innovation. For example, one architectural firm recounted an instance where they refrained from pursuing an innovative solution due to concerns about potential delays caused by seeking dispensation. They feared that the building surveyors might not grant dispensation for the proposed innovative solution, which would have otherwise enhanced accessibility levels. Without dispensation, they would not have achieved the expected points for accessibility. However, they lacked the time to pursue alternative approaches or to redesign the project.

### *4.5 The missing mobilisation*

The final moment of translation is mobilisation, or the process of ensuring that the interests of wider groups of actors are acted upon and the network continues to function. Alliances with competing elements and their networks dominated. Lack of knowledge, accessibility, and its low status together with economy and the understanding of expensive solutions prevented the architectural firms and building surveyors from enrollment. Hence the network was never stabilised and the mobilisation failed to happen and instead alliances with competing networks dominated. Apparently, the competing networks played together and became powerful. Consequently, the traditional use of BR18 overruled any attempt to innovate regardless of which alliance that was into play. The obstacle-problem of the OPP came into play.

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An aspect of translation involved the identity of the actors and their expectations regarding their role in and interest for the network. It became evident that only certain architectural firms and clients could relate to the expectations from the agency. Their specific identity corresponded with an interest in UD and innovation, and they tried to live up to it being aligned to the network in the first place but were hindered. However, other architectural firms did not subscribe to this interest and neither did the clients. Similarly, the interest and the identity of the building surveyors varied. UD was not a part of their identity primarily revolved accessibility and the prescriptive requirements of BR18.

## 5. Discussion

The findings revealed that the translation failed because the actors were not enrolled and locked into the network. We will commence the discussion with a reflection on whether the failed enrollment and mobilisation resulted from a lack of interessement devices or because of powerful competing network? Then follows a discussion about the implications for future adoption of UD in legislation.

### *5.1 Reflection on the failed translation*

The agency failed in preventing the actors of the network from being a part of other networks. Their interessement devices were not strong enough to prevent other alliances, create enrolment in the premise about UD and innovation, and shape the mobilisation. It was not sufficient to believe that the building sector would discover an unclear description of a possibility for innovation and UD presented in a quite unknown guideline instead of chapter 1, which already contained all the regulatory procedures. Likewise, the two ambiguous object clauses were not robust enough as policy tool when the rest of the requirements were prescriptive. Finaly, knowledge would not automatically be absorbed and used, especially not by actors not interested in UD. The understanding of these interessement devices could be characterised as a kind of determinism; as if the possibility was just mentioned in any kind of documents, then everybody would comply with it. The situation was much more complex than that, as the actors were already enrolled and part of other networks that constituted a practice with relations and interactions that were more powerful. The focus on costs and a mindset based on accessibility that furthermore was understood as being expensive and given a low priority together with a lack of knowledge created a combination that was impossible to brush aside because it was the existing practice built upon the prescriptive requirements.

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This regulatory initiative is part of a broader trend towards deregulation where the sector itself is expected to take part in its development (Gottlieb and Frederiksen, 2020). One key insight from this study, which could be seen as a strategy transferable to systems in other countries, is the need for awareness of the complexity of the existing system. Meacham et al (2017) developed a framework for risk-informed performance-based building regulations. Understanding the legal culture, the current state of knowledge, and how the marked embraces this complexity should be considered essential recommendations for policymakers. Our findings indicate that is not merely a matter of describing performance-based criteria. Even in a country like England, which offers the possibility of alternative solutions through an Access Strategy, the situation is far from straightforward. Despite the flexibility and room for manoeuvre, some architectural firms are reluctant to take advantage of these alternatives due to the fear of not obtaining a building permit and being required to redesign. Clients are typically unwilling to cover the costs of a redesign (Grangaard *et al*, 2016). Lack of interest and enrollment can also be a significant issue in contexts with more formalised performance-based systems. Another recommendation would be to focus on involving all stakeholders in the development of initiatives.

### *5.2 Implications for the adoption of UD in regulation*

In this part of the discussion, we will point out the implications of the failed translation in relation to the future adoption of UD. The introduction outlines various explanations for the slow adoption of UD. Nevertheless, the literature is characterized by a tendency to treat these explanations in isolation. In contrast, the findings show that innovation of UD was much more complex, as interconnected issues are related to the capacity of an actor. In other words, an actor could not decide to enroll to the network ignoring the other networks, as they would always be into play. However, Heylighen *et al* (2017) are aware of the multiplicity of explanations, but do not acknowledge power as a factor practiced in the way actors relate to each other and interact.
In contrast, we have shown that there are different facets to the concerns of all actors to make it all fit together when acting in their practice. This means that all the different concerns would be into play all time, but the most powerful will trump the others. Compliance with the prescriptive requirements trumps.

The findings also illustrate that a lack of knowledge was experienced as a hindrance for enrollment, in line with Steinfeld (2010) and Ruptash (2013). However, what Steinfeld and Ruptash do not address is that the existence of knowledge and dissemination of knowledge is not a guarantee that the knowledge is brought

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into play. Knowledge was one of the interessement devices, but it was not strong enough in relation to other alliances. The implications for the dissemination and adoption of UD through policymaking and regulation are that complexity must be acknowledged. This implies a powerful policy that can prevent actors from making alliances with other networks. We know from both the literature (e.g. Grangaard *et al*, 2016) and the findings that clients focus on economy, and that compliance with the buildings regulations is something the clients respect or accept because they must. Thus, an effort to the adoption of UD could be through a fully implementation in policy and regulation.

### *5.3 A performance-based approach to UD*

Imrie (2012) raises the question of whether the dissemination of UD, understood as a process, can support a political change. The perspective of UD as a process has been suggested by others (Steinfeld and Maisel, 2012; Story *et al*, 1998). In this section, we will discuss how a regulatory initiative can be grounded in UD as a process, drawing on examples from the context of Danish fire safety where prescriptive requirements were maintained until 2004 due to a lack of knowledge about fire behaviour (Sørensen, 2014). Today, the system is performance-based and has a greater focus on innovation compared to accessibility and UD.

Three essential elements of performance-based building regulations are goals, performance criteria, and verification methods (Meacham et al, 2005). In Denmark, fire safety is structured around categories of use and risk classes. Categories of use harmonise with UD in relation to the physical, sensory and cognitive abilities of the users, as well as their familiarity with the building. However, a barrier could be a limited understanding of the users. An initiative must foster awareness of all users, not only just those whom the involved actors typically envision (e.g. patients or guests, Grangaard, 2016), but also individuals such as people with disabilities working backstage or on the stage, emphasising all forms of human diversity. The classification of risk can be difficult to translate into UD, as it often focuses on factors like the number of storeys in relation to escape routes. However, instead of concentrating solely on risk, the perspective could be reversed: rather than focusing on escape possibilities, it could centre on access and participation in ways that support health and social integration.

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Defining goals and performance criteria for access and participation might be challenging, as it requires a shift in mindset from prescriptive requirements to more inclusive thinking. In the context of fire safety, it took time for practitioners to adapt to a new way of thinking and working, but this shift opened up different architectural possibilities (Kirkeby *et al*, 2014). A fire strategy report and independent inspection of documentation are tools for verification. A potential barrier to applying a similar approach to UD is that it requires both the knowledge and competences to design and verify. Currently, there is no demand for this. However, just as the field of fire-safety engineering emerged due to performance-based regulation, a similar development could occur with UD, leading to the establishment of a new field with trained specialists, certified experts and a formalised system requiring this expertise. In Sweden, for instance, there are certified specialists in accessibility, and a similar certification initiative could be relevant in relation to performance-based regulation of UD. In addition to a certification system, a UD strategy could also include an economic aspect, contributing to a future body of knowledge about the cost of UD. Consequently, the performance-based approach could advance the status of UD. Future research is warranted, e.g. it would be relevant to study how the new performance based standard EN 17210:2021 will be embraced by clients and architectural firms.

## 6. Conclusion

In this paper, we have provided an understanding of the complexity surrounding a regulatory initiative for innovation in relation to UD. Addressing the first research question about which actors were intended to be enrolled in the network of the initiative about UD innovation, the findings reveal that some architectural firms interested in UD were initially involved in developing innovative solutions and were on the right track towards the network. However, this engagement was short-lived as the network did not manage to keep them enrolled. On the contrary, they experienced that the dialogue with the building surveyors obstructed their design proposals. In contrast, the building surveyor accuse the architectural firms of not being innovative. Moreover, some building surveyors displayed a lack of knowledge about UD, impeding their involvement.

In response to the second research question about what factors contributed to the failure of the mobilisation, the findings suggest that lack of knowledge about accessibility, and its low status together with economic concerns, and the anticipation of costly solutions prevented architectural firms and building surveyors from

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enrollment. Consequently, the network never stabilised and caused the mobilisation to fail. Instead, alliances with competing networks dominated, ultimately favorising traditional requirements of BR18. The agency's efforts to prevent actors from joining other networks were ineffective due to weak interessement devices. The unclear presentation of innovation possibilities in an unfamiliar guideline failed to attract actors already engaged in prescriptive procedures. Economic concerns, low accessibility priorities, and lack of knowledge reinforced existing practices, hindered change.

The study could have also addressed the perspective of the clients. Consequently, we could have delved deeper into their experiences and interest in innovative solutions. Nevertheless, based on the literature, we knew that they only adhere to BR18 and nothing more, which is one of the reasons for not interviewing them. The other reason was that we did not have to capacity to engage with them without deprioritising the architectural firms and building surveyors.

To conclude, the paper demonstrates that the most powerful factor supersedes the others. Compliance with prescriptive requirements takes precedence over other considerations, as observed by Steinfeld (2010) and Ruptash (2013). Lack of knowledge impedes enrollment, but dissemination alone does not guarantee utilisation. Policymaking must address these complexities to effectively promote UD adoption. Clients prioritise economy and compliance, indicating a necessity for comprehensive policy changes away from prescriptive requirements towards UD objectives.

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References

Akrich, M. (1992), “The De-scription of Technical Objects”. Bijker, W. and Law, J. (Ed.s.) *Shaping Technology/Building Society. Studies in Sociotechnical Change*, Cambridge: MIT Press, pp. 205 - 224.

Akrich, M. and Latour, B. (1992), “A summary of a convenient vocabulary for the semiotics human

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This means that anyone may distribute, adapt, and build upon the work for non-commercial purposes, subject to full attribution.
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and nonhuman assemblies", Bijker, W. and Law, J. (Ed.s), *Studies in Sociotechnical Change*. Cambridge: MIT Press, pp. 259 - 265.

Akrich, M., Callon, M., Latour, B. and Monaghan, A. (2002), "The key to success in innovation part I: the art of interessement", *International Journal of Innovation Management*, Vol. 6 No. 2, pp. 187–206.

BR18, Building Regulations (2018), available at https://bygningsreglementet.dk/~/media/Br/BR-English/BR18_Executive_order_on_building_regulations_2018.pdf (accessed 17 April 2024)

Byggeloven (Building Act) (1972), available at https://historisk.bygningsreglementet.dk/file/16082/lovbekg1972.pdf (accessed 17 February 2024)

Bygningsreglementet, (1977), available at https://w2l.sbst.dk/file/502102/br_syvoghalvfjerds.pdf (accessed 17 April 2024)

Callon, M. (1986), "Some elements of a sociology of translation: domestication of the scallops and the fishermen of St Brieuc Bay", Law, J. (Ed.), *Power, action and belief: a new sociology of knowledge?*, Routledge, London, UK, pp. 196–223.

Erdtman, E., Rassmus-Gröhn, K. and Hedvall, P. O. (2021), "Universal design as guiding, striving and unifying: a qualitative study about how universal design is understood, practised and realised in contemporary Sweden", *Scandinavian Journal of Disability Research*, Vol. 23 No. 1, 158-168. doi.org/10.16993/sjdr.770

Flyvbjerg, B. (2006), "Five misunderstandings about case-study research", *Qualitative Inquiry*, Vol. 12 No 2, pp. 219–245. doi.org/10.1177/1077800405284363

Gottlieb, S. C. and Frederiksen, N. (2020), Deregulation as socio-spatial transformation: Dimensions and consequences of shifting governmentalities in the Danish construction industry, *Environment and Planning C: Politics and Space*, Vol. 38 No. 3, pp. 484–502. doi.org/10.1177/2399654419868465

Gramkow, M. C., Merit, M. T. and Stigsdotter, U. K. (2022), A qualitative study on how Danish landscape architectural firms understand and work with accessibility, *Archnet-IJAR: International Journal of Architectural Research*, Vol. 16 No. 3, pp. 536-553. doi.org/10.1108/ARCH-08-2021-0233

Grangaard, S. (2016), "Towards Innovative and Inclusive Architecture", in Lloyd, P. and Bohemia, E (Ed.s.), *Proceedings of DRS2016: Design + Research + Society - Future-Focused Thinking.* Design Research Society, Brighton, United Kingdom, Vol. 8, pp. 3393-3405 doi.org/10.21606/drs.2016.70

Grangaard, S. (2018a), "Clients´Approach to Universal Design – a Slow Change?", in Craddock, G., Doran, C ., McNutt, L. and Rice, D. (Ed.s), *Transforming our World Through Design, Diversity and Education: Proceedings of Universal Design and Higher Education in Transformation Congress 2018.* IOS Press, Studies in Health Technology and Informatics, Vol. 256 pp. 706-715, doi.org/10.3233/978-1-61499-923-2-706

Grangaard, S. (2018b), "A Status of Universal Design in Danish Architectural Policies", in Craddock, G., Doran, C ., McNutt, L. and Rice, D. (Ed.s), , *Transforming our World Through Design, Diversity and Education: Proceedings of Universal Design and Higher Education in Transformation Congress 2018.* IOS Press, Studies in Health Technology and Informatics, Vol. 256 pp. 696-705, doi.org/10.3233/978-1-61499-923-2-696

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Grangaard, S. (2021), "How to Communicate Universal Design to Architects on a New Website? A Reflection on the Type of Knowledge Requested", in Verma, I. (Ed.), *Universal Design 2021: From Special to Mainstream Solutions,* IOS Press, Vol. 282 pp. 301-314. doi.org/10.3233/SHTI210406

Grangaard, S., Frandsen, A. K., & Ryhl, C. (2016). *Analyse af de gældende regler om tilgængelighed i Bygningsreglementet*. SBi Forlag. SBI Vol 2016 No. 31.

Grangaard, S., & Gottlieb, S. (2019), "Opening the Black Box of Accessibility Regulation", in Lill, I. & Witt, E. (Ed.s), *10th Nordic Conference on Construction Economics and Organization,* Emerald Group Publishing, Vol. 2, pp. 365-370. doi.org/10.1108/S2516-285320190000002013

Gray, D. B., Gould, M. and Bickenbach, J. E. (2003), Environmental barriers and disability, *Journal of architectural and planning research*, Vol. 20 No. 1, pp. 29-37.

Guideline BR18. Available at https://bygningsreglementet.dk/Tekniske-bestemmelser/02/Vejledninger/Vejledning-til-brugerbegreb (accessed 17 February 2024)

Heylighen, A., Van der Linden, V. and Van Steenwinkel, I. (2017), Ten questions concerning inclusive design of the built environment. *Building and environment*, Vol.114, pp. 507-517. doi.org/10.1016/j.buildenv.2016.12.008

Ielegems, E., Herssens, J., Nuyts, E. and Vanrie, J. (2019), DRIVERS AND BARRIERS FOR UNIVERSAL DESIGNING, *Journal of Architectural and Planning Research*, Vol. 36 No 3, pp.181-197.

Imrie, R., and Street, E. (2011), *Architectural design and regulation*. John Wiley & Sons.

Imrie, R. (2012), Universalism, universal design and equitable access to the built environment, *Disability and rehabilitation*, Vol. 34 No. 10, pp. 873-882. doi.org/10.3109/09638288.2011.624250

Kirkeby, I. M., Ryhl, C., Frandsen, A. K, and Pedersen, L. S. (2014), *Funktionsbaserede tilgængelighedskrav? Analyse af udfordringer og barrierer for en eventuel ændring af bygningsreglementets detaljerede tilgængelighedskrav til funktionsbaserede krav.* SBI forlag. Vol. 2014 No. 09

Latour, B. and Woolgar, S. (1986), *Laboratory Life: The Construction of Scientific Facts*. 2nd edition. Princeton University Press.

Latour, B. (1987), *Science in action: How to follow scientists and engineers through society*. Harvard university press.

Latour, B. (1992), 'Where are the missing masses? The sociology of a few mundane artifacts', Bijker, W. E. and Law, J. (Ed.s.) *Shaping Technology/Building Society: Studies in Sociotechnical Change*, Cambridge, MA, MIT Press, pp. 225-58.

Latour, B. (2005), *Reassembling the Social. Oxford University Press*, New York.

Law, J. (2008), "Actor network theory and material semiotics", Turner, B.S. (Ed.) *The new Blackwell companion to social theory*, Wiley-Blackwell, Malden, Massachusetts, pp. 141–158.

Lid, I. M. (2013), Developing the theoretical content in Universal Design. *Scandinavian Journal of Disability Research*, Vol. 15 No. 3, pp. 203-215. doi.org/10.1080/15017419.2012.724445

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Meacham, B., Bowen, R., Traw, J. and Moore, A. (2005), Performance-based building regulation: current situation and future needs. *Building Research & Information*, Vol. 33 No. 2, pp. 91-106. doi.org/10.1080/0961321042000322780

Meacham B.J, and van Straalen I.J. (2017), A socio-technical system framework for risk-informed performance-based building regulation. *Building Research & Information*. Vol. 46, NO.4, pp. 444-462. doi.org/10.1080/09613218.2017.1299525

Müller, L., Wojahn, D., Sandström, I., and Hedvall, P. O. (2022), Planning for human diversity: design patterns of Universal Design. *Nordic Journal of Architectural Research*, Vol. 34 No. 2, pp. 75-104.

Regeringen (2013) 'Disability policy action plan, A Society for All'. available at https://sm.dk/media/8317/disability-policy-action-plan-2013-one-society-for-all.pdf (accessed 17 April 2024)

Regeringen, 2014. Vejen til et styrket byggeri i Danmark – regeringens byggepolitiske strategi.

Ruptash, S. (2013), Universal Design through Passion, Knowledge and Regulations?. In *Trends in Universal Design*, pp. 24- 27.

Salmen, J. P. (2011), Universal design for academic facilities. *New Directions for Student Services*, Vol. 2011 No. 134, pp.13-20.

Steinfeld, E. (2010), Advancing Universal Design, in *The State of the Science in Universal Design: Emerging Research and Development*. Buffalo (NY): State University of New York, Bentham eBooks, pp. 1-19.

Steinfeld, E. (2013), Creating an inclusive environment. In *Trends in Universal Design*, pp. *52-57*.

Steinfeld, E. (2023), The space of accessibility and universal design, Lid, I. M., Steinfeld, E., and Rembis, M. (Ed.s.), *Rethinking Disability and Human Rights: Participation, Equality and Citizenship,* Routledge, New York, USA, pp. 100-118. DOI: 10.4324/9781003052708-11

Steinfeld, E., and Maisel, J. (2012), *Universal design: Creating inclusive environments*. John Wiley & Sons.

Steinfeld, E., and Tauke, B. (2002), 2.1 Universal Designing, Christophersen, J (Ed.), *17 Ways of Thinking and Teaching*, pp. 165-189.

Story M.F., Mueller J.L. and Mace R.L (1998), *The universal design file: Designing for people of all ages and abilities*. Raleigh: NC State University.

Sørensen, L.S. (2014), *Fire-safety engineering and performance-based codes*. Lyngby: Polyteknisk Forlag.

Van der Linden, V., Dong, H. and Heylighen, A. (2016), From accessibility to experience: Opportunities for inclusive design in architectural practice. *Nordic Journal of Architectural Research*, Vol. *28* No. 2, pp.33-55

Wauters, H., Vermeersch, P.W. and Heylighen, A. (2014), Reality check: notions of accessibility in today's architectural design practice, in Lim, Y.-K., Niedderer, K., Redström, J. Stolterman, E. and Valtonen, A. (Ed.s.), *DRS 2014: Design's Big Debates*. Umeå, Design Research Society & Umeå Institute of Design, pp. 1482-1491.

Yusof, M., and Jones, D. (2014), Universal design practice in Malaysia: Architect's perceptions of its terminology. Proceeding of Universal Design 2014: Three Days of Creativity and Diversity

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Caltenco, H., Hedvall, P.O., Larsson, A., Rassmus-Gröhn, K. and Rydeman, B. (Ed.s.) *Assistive Technology Research Series*, Vol. 35, pp. 347-355. Doi:10.3233/978-1-61499- 403-9-347

Zallio, M., and Clarkson, P. J. (2021), Inclusion, diversity, equity and accessibility in the built environment: A study of architectural design practice. *Building and Environment*, Vol. 206, December 2021, 108352. https://doi.org/10.1016/j.buildenv.2021.108352